

Referee Report: "Anchored Abroad: Sunk Costs, Asset Specificity, and Japanese Affiliate Exit from Brexit Britain"

Summary

This paper investigates changes in Japanese FDI into the UK after the 2016 Brexit referendum, which increased uncertainty about the future UK-EU economic relationships. To do so, the authors set up a stylized conceptual framework to show how entry sunk costs can affect the decision to enter a market and cause hysteresis. Using Japanese data, which registers information about Japanese parent firms and their affiliates abroad, they estimate a battery of firm-level diff-in-diff and host country-level synthetic diff-in-diff models. The main result is that there is no significant impact on Japanese affiliates' exit from the UK after the referendum relative to other European economies. They also show that industries with low asset specificity did experience differential exit, consistent with the fact that they could recover a greater proportion of the investment in these sectors.

Assessment

I enjoyed reading this well-written paper on a matter that I think is first-order in the current global economic policy landscape. Micro evidence on the impact of economic disintegration on FDI is surprisingly scarce in the literature. Given this and the interesting data and empirical strategy employed, I believe the paper is well-positioned to make a significant contribution to the literature. Having said so, I have many comments that would increase confidence in the results and highlight their significance across the board. In this sense, the current draft, in my view, is lacking and would greatly benefit from a revision. The comments below aim to assist in this regard (in no particular order).

Comments

1. **Not the actual Brexit.** The paper uses phrases such as “the impact of Brexit on...” repeatedly, even in the abstract, but the study is not about Brexit per se as we generally understand it. Brexit factually occurred in 2021, and the analysis here is centered on the 2016 referendum. Firms in the UK could trade freely with the EU throughout the period analyzed, and they were under the same EU regulations as before 2016. What the authors may actually be capturing is the impact of uncertainty about the timing and actual form of Brexit once it was clear that the UK was going to leave the EU. I think the authors should be more precise with this. In addition, this may explain the weak results estimated for this period.
2. **Order and Relevance of Results.** The paper is organized in a way that focuses on the “null” results. It begins with estimates that pool over sectors and shows, using a variety of estimators and robustness checks, that the effect of “Brexit” on Japanese affiliates' exit, although positive, is insignificant. This result is indeed interesting, and I am all about increasing the number of “null” results that are published. However, the paper shows towards the end what I think is the most interesting result: asset specificity matters. In my

view, the pooled vs asset specificity heterogeneity should be the first result, and then I would focus on showing that “Brexit” did have a robust impact on exit in these sectors.

3. **Pooled Results.** All the estimations compute year-specific results. This implies that the estimates that are analyzed are to be interpreted as year-specific differences versus the base period. This is fine, but it does not speak to the average post-treatment effect of “Brexit”. Specifically, the authors should include the typical pooled diff-in-diff results too, on top of the event studies. This means estimating a coefficient by pooling the 2016-2020 effect (and for which the authors will have more statistical power). If the authors pursue this, they should be careful not to mix yearly and bi-yearly data, as the growth rates will be mixed up. In other words, they should include only 2010, 2012, 2014, 2016, 2018 and 2020.
4. **EU as a Control Group.** Are other EU countries a good control group for the UK? My conjecture is that they are not because many Japanese firms seem to have chosen the UK to serve other EU countries when Brexit was not even in the dictionary (the UK had a disproportional share of affiliates relative to its relative EU size). If this is the case, some firms may have “upped sticks” and settled in other EU countries (e.g. see the UK and Germany curves post-2018 in Figure 1). I suggest including results in which non-EU high-income countries are included, and in which both EU and non-EU high-income countries are included.
5. **UK importance for Japan.** It is clear that the UK is important for Japanese multinationals within the EU. Is it important for their entire operations? The authors seem to be in a good position to check this and investigate a further heterogeneous effect depending on the importance of the UK. I suspect that firms with a small share of involvement in the UK in terms of their overall activity may have found it easier to pull out of the UK.
6. **Sector variation.** The authors seem to have the information about the affiliates’ sectors (or parents’). Different sectors were exposed differently to Brexit. Within goods, some sectors faced a potentially higher tariff in the event of Brexit (e.g. Graziano et al., 2021). Within services, they faced different regulations (e.g. Ahmad et al., 2022).¹ I think employing “Brexit-intensity” variation across sectors to identify the effect on exit would strengthen the paper.
7. **Firm vs Aggregate Effects.** Most estimates are done at the affiliate level. This implies that if most firms do not exit relative to the control group, the effect is likely to be insignificant. However, the ones that exit may be the larger companies, causing a more significant economic impact in both the UK and Japan. The opposite may be the case, too. I suggest the two different strategies to tackle these types of compositional effects:
 - a. **Sectoral level aggregation.** Count affiliates at the sectoral level and estimate a PPML regression with the count on the LHS. It would be helpful to utilize the sectoral variation mentioned above.

¹ Ahmad, S., Limão, N., Oliver, S., & Shikher, S. (2023). Brexit uncertainty and its (dis) service effects. *American Economic Journal: Economic Policy*, 15(4), 459-485, and Graziano, A. G., Handley, K., & Limão, N. (2021). Brexit uncertainty and trade disintegration. *The Economic Journal*, 131(635), 1150-1185.

b. Weight by employment. Weigh either (or both) firms or sectors by their employment level.

8. Model with Divestiture. The model is conceptually neat and fulfills its role of providing a framework for the empirics. However, I think it misses the opportunity to study this aspect of FDI a little deeper. Specifically, I think it should consider divestiture as a choice variable for the firm. Some comparative statics changing the entry and exit cost and the probability of changing economic and policy conditions would be great to add further relevance to this section.

9. Robustness to Specific Countries. The analysis is done by using high-income EU countries. Given that they are just a few, the authors should check if the results are robust to leaving one out alternatively. In other words, is there one specific country driving the results? There is nothing particularly wrong with this, in principle, but it should be shown.

10. Compare Entry vs Exit using Mutually Exclusive Samples. The entry results are relegated to the last subsection and are computed using the SDiD method. I think analyzing the impact of “Brexit” on entry is equally important as analysing its impact on exit. Moreover, it is also given similar importance in the model section. What can be done is to use the sample of firms that are not present in 2010 (i.e. the complement of the exit sample) and estimate the impact on this sample. Provided the assumption that these firms were “new” after 2010, the estimate could be interpreted as “new entry” and compared with the exit coefficient. In principle, I would expect a stronger impact on new entry compared to exit in absolute value.

11. Other comments.

- a. In the preliminary analysis, the left and right graphs in Figure 1 vary in two aspects: the number of countries and levels versus growth, therefore making them not directly comparable. I suggest only changing one dimension and relegating the other to the appendix.
- b. All the graphs summarizing the estimations miss key information about the estimations performed: number of observations, R^2 , etc. I suggest including corresponding tables with the point estimates, standard errors, etc., in the appendix.
- c. What is the share of firms and observations that are kept in the sample by choosing affiliates present in 2010? How much are we missing of the total? I could not find this first-order information easily.
- d. Employment is used as a control later in the paper, and it requires imputation. How much imputation is done for employment in terms of observations? Is it possible to also present the results excluding observations without employment?
- e. Please include an Appendix table showing the weights computed for the SDiD.